One-pot PCR mixture and conditions (head-tailed gene-region primers + barcoded-head primers in a single reaction)

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Lab plain-PDF edition, DRAFT v0.1 — written 2026-10-20, not yet bench-validated

Source of record: HuangLab (Yin-Tse Huang lab, NCKU). Hand-written lab protocol, derived from the lab's two-step PCR protocol (*2-step PCR mixture and conditions, barcoded-head primers*) and collapsed into one tube. Companion workflow chart: *One-pot PCR amplicon library workflow* on the lab protocol page.

> **Status: DRAFT v0.1.** Reagent volumes follow the lab's two-step protocol. The one-pot-specific values (gene-primer : barcode-primer ratio, phase split of the cycling program) are literature-typical starting values and **must be validated** against the two-step product on at least 3 known templates before this protocol replaces the two-step route for a project. Record the validation run in the LIMS and bump the version here.

Abstract

Amplify a gene region and attach the Nanopore barcode in **one reaction**. The head-tailed gene-region primers are kept **limiting** (they are consumed in the early cycles) and the barcoded-head primers are in **excess** (they take over in the later cycles by priming on the head sequence). The product is the same molecule the two-step route makes: *barcode – head – gene region – head – barcode*. Each sample therefore gets its own tube with its own barcode, and tubes are pooled equimolar afterwards.

When to use one-pot instead of two-step

- Many samples, same gene region, templates of good quality (clean fungal or bacterial DNA, or soil extracts that already amplify well in the two-step route).
- You want to halve hands-on time and the number of opportunities for cross-contamination.
- Do **not** use one-pot for difficult templates (inhibitor-rich soil, low-copy targets) or when a gel-checkable intermediate is needed: there is no 1st-PCR product to inspect.

Materials

1. 2X KAPA HiFi HotStart ReadyMix (LIMS `taq` default: `KAPA HiFi HotStart`)
2. Head-tailed gene-region primers, forward + reverse, 10 µM working stocks (same primers as the 1st PCR of the two-step protocol; refer to the lab primer list for the region and its annealing temperature)
3. Pre-mixed barcoded-head primer pairs (forward + reverse), 10 µM each, one pair per barcode, in the barcode plate used for the 2nd PCR of the two-step protocol
4. Nuclease-free water (ddH2O)
5. DNA template, normalised where possible (1–10 ng/µL); MolWork# assigned
6. 8-strip PCR tubes or 96-well PCR plate, filter tips, 1% bleach for the bench

> **Note**
>
> Barcoded-head primers must carry the **same head sequence** as the tail of the gene-region primers; otherwise the second phase cannot prime. Check the primer list before mixing a new region with the barcode plate.

Reaction mixture (per 25 µL reaction)

Prepare the master mix **without barcode primers**, dispense, then add the barcode pair and the template per tube. Prepare 1.2X of the master mix volume for pipetting loss. **A negative control (no template) is always included**, carrying its own barcode.

Component	Volume	Final conc.	Role
2X KAPA HiFi HotStart ReadyMix	12.5 µL	1X	polymerase + buffer + dNTPs
Head-tailed gene-region primer F (10 µM)	0.125 µL	**0.05 µM**	limiting, phase 1
Head-tailed gene-region primer R (10 µM)	0.125 µL	**0.05 µM**	limiting, phase 1
ddH2O	9.75 µL	—	
Master mix subtotal	*22.5 µL*		
Pre-mixed barcoded-head primers F+R (10 µM each)	1.25 µL	**0.5 µM each**	excess, phase 2
DNA template	1.25 µL	1–10 ng per reaction	
Total	**25 µL**		

> **Note**
>
> The 0.05 µM : 0.5 µM ratio (1 : 10) is the starting point. If the gel shows a strong head-only band (short product, no barcode) raise the barcode primers to 0.75 µM or lower the gene primers to 0.025 µM. If yield is poor, raise the gene primers to 0.1 µM and add 2–3 cycles to phase 1.

> **Note**
>
> 0.125 µL pipetting is unreliable. Make a **1 µM working dilution** of each gene-region primer and add 1.25 µL of it (then reduce ddH2O to 7.5 µL per reaction).

Procedure

Setup

Step 1

Wear gloves. Clean the bench with 1% bleach. Thaw the ReadyMix, primers and templates on ice; vortex briefly and spin down.

Step 2

Prepare the master mix (ReadyMix + gene-region primers + ddH2O) for n + 1 reactions × 1.2. Mix gently by pipetting; quick spin.

Step 3

Dispense **22.5 µL** master mix into each PCR tube / well.

Step 4

Add **1.25 µL** of the pre-mixed barcoded-head primer pair to each tube, **one barcode per tube**. Write the barcode ID on the tube and in the LIMS PCR row before continuing.

Step 5

Add **1.25 µL** DNA template to each tube. The negative control receives 1.25 µL ddH2O instead.

Step 6

Mix gently by tapping, quick spin. Keep on ice until the cycler is ready.

Cycling program (two-phase, single run)

Step 7

Run the program below. Phase 1 anneals at the **gene-region primer** temperature; phase 2 anneals at the **head** temperature, where only the barcoded-head primers prime efficiently.

Phase	Step	Temp	Time	Cycles
—	Initial denaturation	95 °C	3 min	1
1 (gene region)	Denaturation	98 °C	20 s	12 cycles
1	Annealing	**60 / 62 / 64 °C** (a)	15 s	
1	Extension	72 °C	1 min per kb (b)	
2 (barcoding)	Denaturation	98 °C	20 s	20 cycles
2	Annealing	65 °C (c)	15 s	
2	Extension	72 °C	1 min per kb (b)	
—	Final extension	72 °C	5 min	1
—	Hold	4 °C	∞	

(a) Choose the phase-1 annealing temperature from the gene-region primer list; the LIMS `method` value records it as `one_pot_60`, `one_pot_62` or `one_pot_64`. (b) Use the extension time of the two-step protocol for the same region (e.g. 3 min for ~2–3 kb products). (c) 65 °C is the validated 2nd-PCR annealing temperature of the two-step protocol; the head sequence does not change between regions.

> **Note**
>
> Total 32 cycles. Do not extend phase 2 beyond 25 cycles: over-cycling increases chimeras and barcode-primer dimers. If the product is weak, add cycles to phase 1 first.

Quality control

Step 8

Run **2 µL** of each reaction on a 1% agarose gel next to the ladder. Expected product = gene-region amplicon + ~2 × (head + barcode). One band per lane; the negative control must be blank.

Observation	Interpretation	Action
Single band at expected size	one-pot worked	continue
Two bands: expected + smaller (head-only)	phase 2 incomplete	raise barcode primers / phase-2 cycles
Smear or band below 150 bp	primer dimer	lower barcode primers; check template amount
Band in negative control	contamination	discard the batch, re-clean the bench

Step 9

Photograph the gel, mark the wells, upload to the lab Google Drive, and record `pcr_ok` for each row.

Step 10

Quantify each tube on the **Qubit Flex** (dsDNA HS). Record the concentration in the LIMS; this value drives the equal-molar pooling.

LIMS registration (Stage 3b, `/lims-register`)

Step 11

One `pcr_reaction` row per MolWork#. Set `method` = `one_pot_<Ta>`, `taq` = `KAPA HiFi HotStart`, primers, date, performed_by. Because the barcode is already in the single reaction, assign the **flow**

cell and barcode at setup and take the single OK: the grid mirrors `pcr_ok` → `second_pcr_ok` for `one_pot_*` rows, so the sample becomes ready for Stage 4a as soon as the gel passes.

Downstream

Step 12

Pool all barcoded tubes **equal molar**, then purify the pool with the *Size selection (Purification) (1X ratio)* protocol and proceed to library construction (EvoPrep). See the one-pot workflow chart for the full path.

Validation checklist (before promoting this draft to v1.0)

- ☐ 3 known templates amplified by one-pot and by two-step in parallel; band sizes match.
- ☐ Negative control blank in both routes.
- ☐ Equimolar pool sequenced; per-barcode read counts within 3-fold of each other.
- ☐ Reads classify to the expected taxa; chimera rate not higher than the two-step run.
- ☐ Record the run (Nanopore#), the primer ratio used and the annealing temperature here, and bump to v1.0.